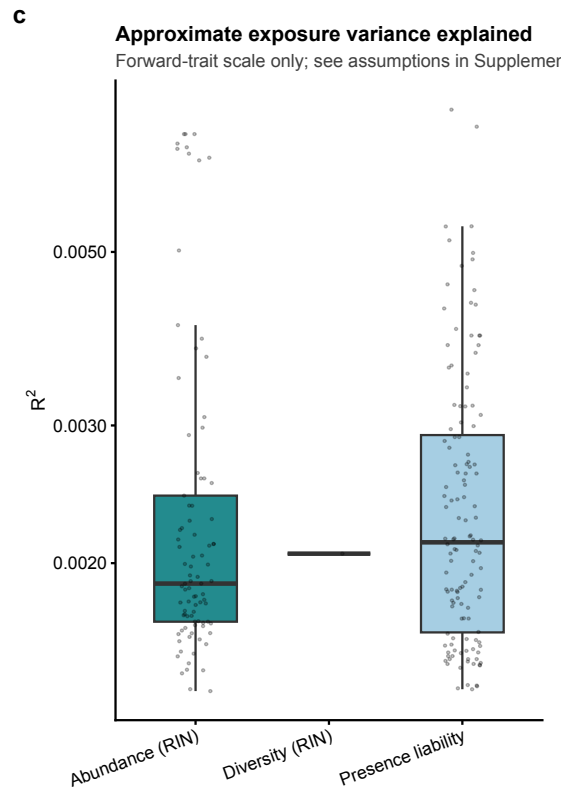
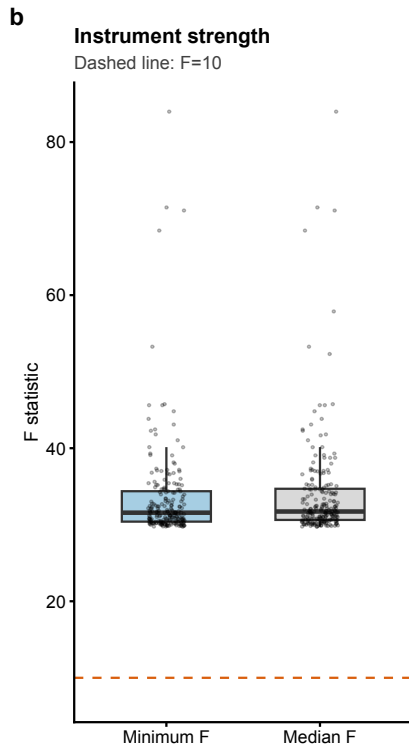
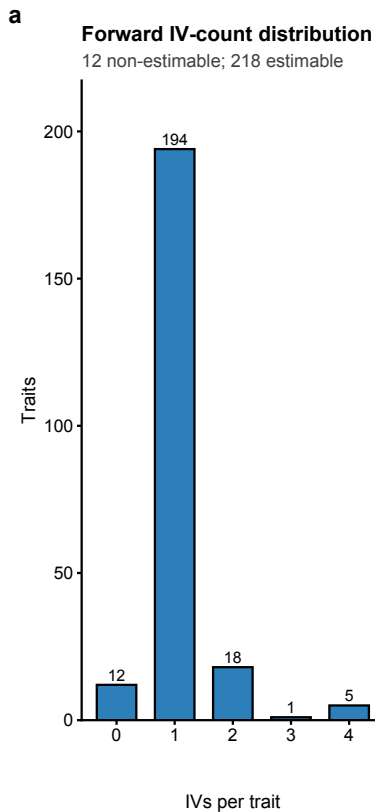
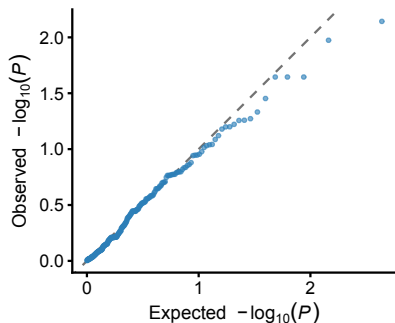
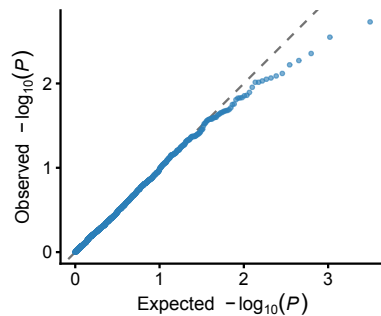




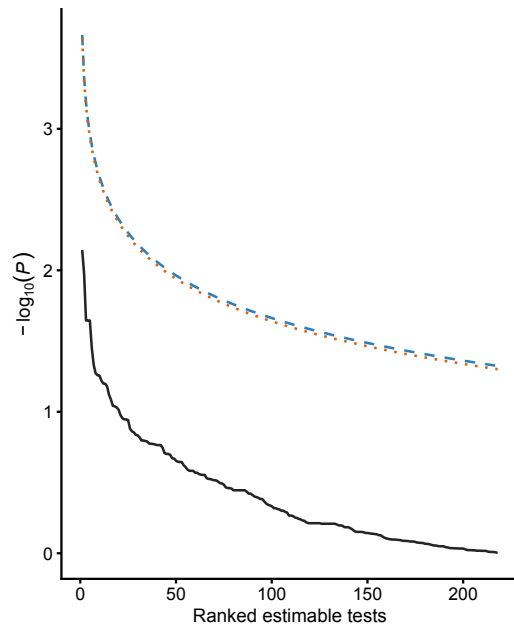
a Forward (218 estimable)



b Reverse (1 572 estimable)



c Forward BH sensitivity
0 FDR signals in all corrections
Unique clusters n=225; minimum q=0.893–0.961

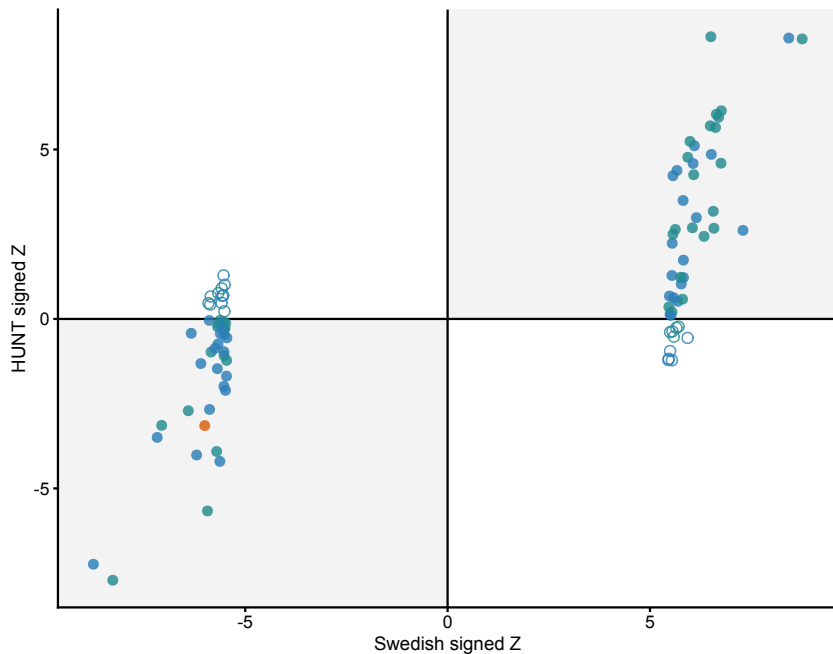


— Observed - - - Predefined family n=230 . . . Estimable only n=218

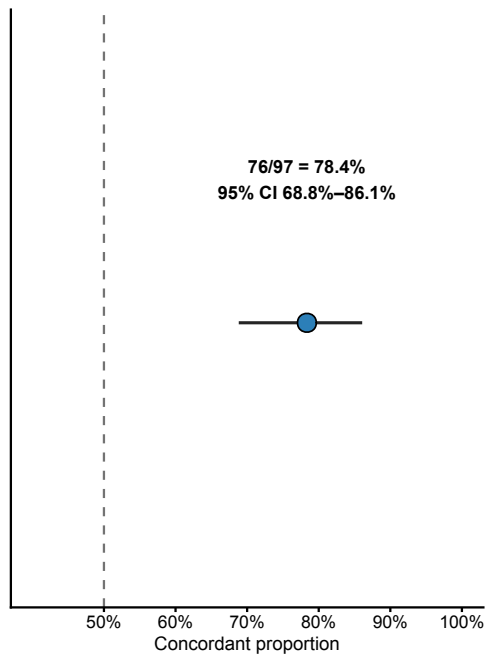
QQ plots are descriptive because microbial traits are correlated; no genomic-inflation estimate is interpreted as if tests were independent.

a**Exact-label same-SNP exposure associations**

97/97 exact-label Swedish lead SNPs found; no fuzzy matching

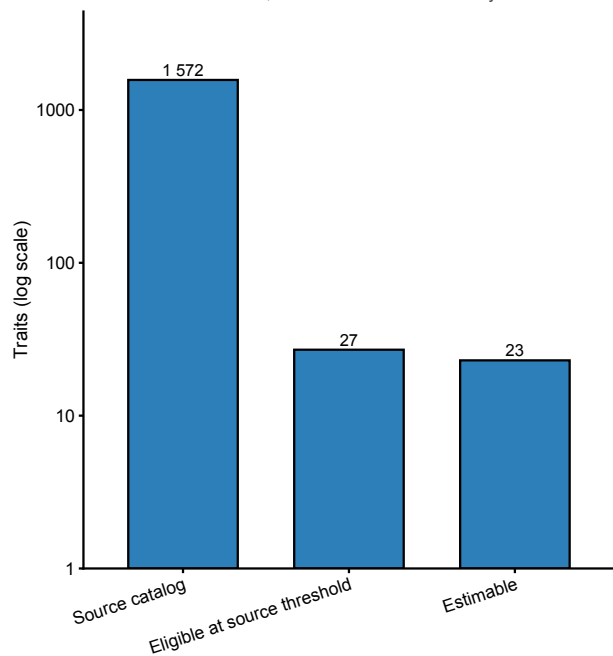
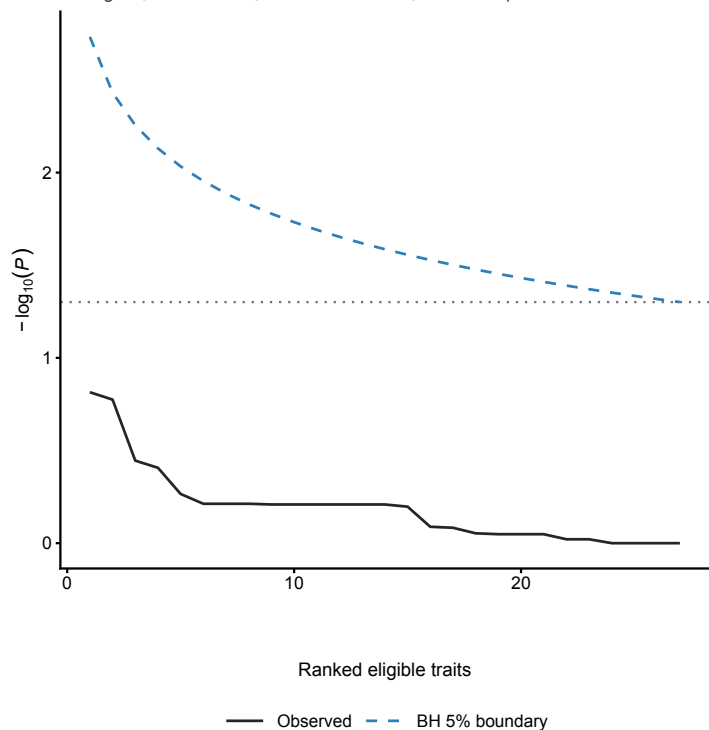
**b****Direction concordance**

Descriptive binomial interval; trait correlation not modelled



○ Direction discordant ● Direction concordant Swedish phenotype ● Swedish abundance ● Swedish diversity ● Swedish presence

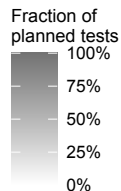
Signed Z supports a direction-only comparison. Swedish presence and HUNT normalized abundance scales preclude direct comparison of raw beta or MR OR magnitude.

a**Source-study-wide threshold eligibility**Thresholds: 1.7×10^{-6} , 5.4×10^{-11} or 4.3×10^{-10} by trait class**b****Strict-threshold MR sensitivity**27 eligible; 23 estimable; minimum $P=0.153$; minimum $q=1$ 

a**Prespecified bounded mechanistic screening**

Counts are shown within complete testing families

Function → ED	5	5	0	0	0.482
Cytokine → ED	40	18	1	0	1.000
Endothelial protein → ED	9	7	0	0	0.395
Function → cytokine	200	169	7	0	0.946
Function → endothelial protein	45	45	2	0	0.489
Cytokine indirect effect	200	80	0	0	1.000
Endothelial indirect effect	45	35	0	0	1.000
	Planned	Estimable	Nominal P<0.05	FDR q<0.05	

b**Minimum q**

No family produced an FDR-significant result. The display summarizes evidence counts and does not imply biological pathway support.